# ADVANCED CONTROL SYSTEMS

## Manipulator Dynamics

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# Outline



Kinetic and Potential Energy of a Rigid body

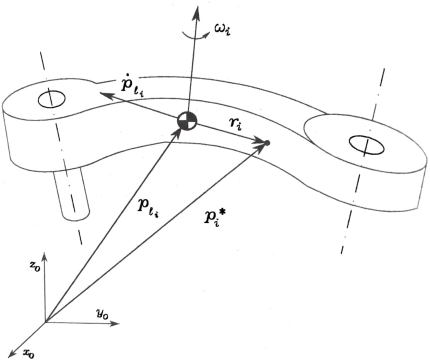
PROJECT

Equations of Motion

PROJECT

# Kinetic and Potential Energy of a Rigid body

Kinematic description of Link *i*




- $\Sigma_0 = \{x_0, y_0, z_0\}$ base reference frame
- $m_i$ mass of link $i$

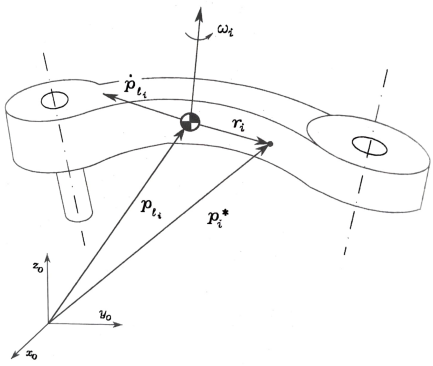
$$m_i = \int_{V_{\ell_i}} \rho dV = \int_{V_{\ell_i}} \rho(x, y, z) dx dy dz$$

- $p_i^*$ position ($\in \mathbb{R}^3$) of the generic point w.r.t. $\Sigma_0$
- $p_{\ell_i}$ position ($\in \mathbb{R}^3$) of the center of mass w.r.t. $\Sigma_0$

$$p_{\ell_i} = \frac{1}{m_{\ell_i}} \int_{V_{\ell_i}} p_i^* \rho dV$$

$$p_i^* = p_{\ell_i} + r_i \qquad r_i = p_i^* - p_{\ell_i}$$

Kinematic description of Link *i*




- $\dot{p}_{\ell_i}$ linear velocity ($\in \mathbb{R}^3$) of the center of mass w.r.t. $\Sigma_0$
- $\omega_i$ angular velocity ($\in \mathbb{R}^3$) of the center of mass w.r.t. $\Sigma_0$
- $\dot{p}_i^*$ linear velocity ($\in \mathbb{R}^3$) of the generic point w.r.t. $\Sigma_0$

$$\begin{aligned} \dot{p}_i^* &= \dot{p}_{\ell_i} + \omega_i \times r_i \\ &= \dot{p}_{\ell_i} + S(\omega_i) r_i \end{aligned}$$

**Assumption 1**: rigid links
**Assumption 2**: rigid transmission

The total kinetic energy is given by the sum of the contributions relative to the motion of each link ($\mathcal{T}_{\ell_i}$) and the contributions relative to the motion of each joint motor actuator ($\mathcal{T}_{m_i}$)

$$\mathcal{T} = \sum_{i=1}^{n} (\mathcal{T}_{\ell_i} + \mathcal{T}_{m_i})$$

*From now on, we will consider only $\mathcal{T}_{\ell_i}$. The equations for $\mathcal{T}_{m_i}$ can be found in the textbook.*

$$\mathcal{T}_{\ell_i} = \frac{1}{2} \int_{V_{\ell_i}} (\dot{p}_i^*)^T \dot{p}_i^* \rho dV$$

$$
\begin{aligned}
\mathcal{T}_{\ell_i} &= \frac{1}{2}\int_{V_{\ell_i}} (\dot{p}_i^*)^T \dot{p}_i^* \rho dV \\
&= \frac{1}{2}\int_{V_{\ell_i}} (\dot{p}_{\ell_i} + S(\omega_i) r_i)^T (\dot{p}_{\ell_i} + S(\omega_i) r_i) \rho dV \\
&= \underbrace{\frac{1}{2}\int_{V_{\ell_i}} \dot{p}_{\ell_i}^T \dot{p}_{\ell_i} \rho dV}_{\textit{Translational}} + \underbrace{\int_{V_{\ell_i}} (\dot{p}_{\ell_i})^T S(\omega_i) r_i \rho dV}_{\textit{Mutual}} + \underbrace{\frac{1}{2}\int_{V_{\ell_i}} r_i^T S(\omega_i)^T S(\omega_i) r_i \rho dV}_{\textit{Rotational}}
\end{aligned}
$$

where the *translational energy* is the kinetic energy of a point mass at CoM

$$
\underbrace{\frac{1}{2}\int_{V_{\ell_i}} \dot{p}_{\ell_i}^T \dot{p}_{\ell_i} \rho dV}_{\textit{Translational}} \overset{(\star)}{=} \dot{p}_{\ell_i}^T \dot{p}_{\ell_i} \frac{1}{2}\int_{V_{\ell_i}} \rho dV = \frac{1}{2} m_{\ell_i} \dot{p}_{\ell_i}^T \dot{p}_{\ell_i}
$$

$(\star)$: $\dot{p}_{\ell_i}$ does not depend on $dV$

The *mutual energy* is equal to zero

$$\underbrace{\int_{V_{\ell_i}} (\dot{p}_{\ell_i})^T S(\omega_i) r_i \rho dV}_{Mutual} = \int_{V_{\ell_i}} (\dot{p}_{\ell_i})^T S(\omega_i)(p_i^* - p_{\ell_i})\rho dV$$

$$\stackrel{(\square)}{=} (\dot{p}_{\ell_i})^T S(\omega_i) \left( \int_{V_{\ell_i}} p_i^* \rho dV - p_{\ell_i} m_{\ell_i} \right)$$

$$= (\dot{p}_{\ell_i})^T S(\omega_i) (p_{\ell_i} m_{\ell_i} - p_{\ell_i} m_{\ell_i})$$

$$= 0$$

($\square$): $\dot{p}_{\ell_i}$ and $\omega_i$ do not depend on $dV$

The *rotational energy*

$$\underbrace{\frac{1}{2}\int_{V_{\ell_i}} r_i^T S(\omega_i)^T S(\omega_i) r_i \rho dV}_{\text{Rotational}} \overset{(\triangle)}{=} \frac{1}{2}\int_{V_{\ell_i}} \omega_i^T S(r_i)^T S(r_i)\omega_i \rho dV$$

$$\overset{(\triangledown)}{=} \frac{1}{2}\omega_i^T \left(\int_{V_{\ell_i}} S(r_i)^T S(r_i)\rho dV\right)\omega_i$$

$$\overset{(\diamondsuit)}{=} \frac{1}{2}\omega_i^T I_{\ell_i}\omega_i$$

$(\triangle)$: $S(\omega_i)r_i = -S(r_i)\omega_i, \qquad S(r_i) = \begin{bmatrix} 0 & -r_{iz} & r_{iy} \\ r_{iz} & 0 & -r_{ix} \\ -r_{iy} & r_{ix} & 0 \end{bmatrix}$

$(\triangledown)$: $\omega_i$ does not depend on $dV$

$(\diamondsuit)$: $I_{\ell_i} \triangleq \int_{V_{\ell_i}} S(r_i)^T S(r_i)\rho dV = \begin{bmatrix} I_{\ell_i xx} & -I_{\ell_i xy} & -I_{\ell_i xz} \\ * & I_{\ell_i yy} & -I_{\ell_i yz} \\ * & * & I_{\ell_i zz} \end{bmatrix}$

# Kinetic Energy

- $I_{\ell_i}$ is the *inertia tensor* relative to the centre of mass of Link *i* expressed in the base frame $\Sigma_0$
- $I_{\ell_i} = I_{\ell_i}^T$, *symmetric matrix*
- $I_{\ell_i}$ depends on $q$, i.e. it is *configuration-dependent*

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What happens to the inertia tensor when expressed w.r.t. the frame $\Sigma_i$ ($I_{\ell_i}^i$) attached to the Link $i$ instead of $\Sigma_0$?

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What happens to the inertia tensor when expressed w.r.t. the frame $\Sigma_i$ ($I_{\ell_i}^i$) attached to the Link $i$ instead of $\Sigma_0$?

Since

$$\frac{1}{2}\omega_i^T I_{\ell_i}\omega_i = \frac{1}{2}(\omega_i^i)^T I_{\ell_i}^i \omega_i^i$$

(i.e. the product is invariant with respect to the chosen reference frame) and exploiting $\omega_i^i = R_i^T \omega_i$, we have

$$I_{\ell_i} = R_i\, I_{\ell_i}^i\, R_i^T \quad , \qquad (\, I_{\ell_i}^i = R_i^T\, I_{\ell_i}\, R_i\,)$$

# Kinetic Energy

- $I_{\ell_i}(q) = R_i(q)\, I^i_{\ell_i}\, R_i^T(q)$
- $I^i_{\ell_i}$ is constant, *configuration-independent*
- If the axes of Link *i* frame coincide with the central axes of inertia, then the inertia cross-products are null and the inertia tensor relative to the centre of mass is a *diagonal matrix*

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The kinetic energy is

$$\begin{aligned} \mathcal{T}_{\ell_i} &= \frac{1}{2} m_{\ell_i} \dot{p}^T_{\ell_i} \dot{p}_{\ell_i} + \frac{1}{2} \omega_i^T I_{\ell_i} \omega_i \\ &= \frac{1}{2} m_{\ell_i} \dot{p}^T_{\ell_i} \dot{p}_{\ell_i} + \frac{1}{2} \omega_i^T R_i\, I^i_{\ell_i}\, R_i^T \omega_i \end{aligned}$$

where $\dot{p}_{\ell_i}$ and $\omega_i$ are function of $q$ (besides $R_i$, of course)

We actually proved the König's theorem

## Theorem

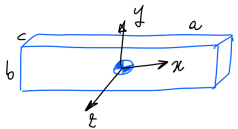
*The kinetic energy of a system of particles is the sum of the kinetic energy associated to the movement of the center of mass ($\star$) and the kinetic energy associated to the movement of the particles relative to the center of mass ($\lozenge$).*

$$\mathcal{T}_{\ell_i} = \underbrace{\frac{1}{2} m_{\ell_i} \dot{p}_{\ell_i}^T \dot{p}_{\ell_i}}_{(\star)} + \underbrace{\frac{1}{2} \omega_i^T R_i \, I_{\ell_i}^i \, R_i^T \omega_i}_{(\lozenge)}$$

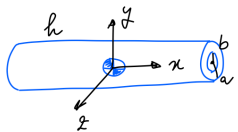
# Examples of body inertia matrices

Assumptions: homogeneous body of mass $m$ with symmetry

Body inertia matrices (w.r.t CoM): $I_C = \begin{bmatrix} I_{C,xx} & 0 & 0 \\ 0 & I_{C,yy} & 0 \\ 0 & 0 & I_{C,zz} \end{bmatrix}$




$$I_C = \begin{bmatrix} \frac{1}{12}m(b^2+c^2) & 0 & 0 \\ 0 & \frac{1}{12}m(a^2+c^2) & 0 \\ 0 & 0 & \frac{1}{12}m(a^2+b^2) \end{bmatrix}$$




$$I_C = \begin{bmatrix} \frac{1}{2}m(a^2+b^2) & 0 & 0 \\ 0 & \frac{1}{12}m\left(3(a^2+b^2)+h^2\right) & 0 \\ 0 & 0 & \frac{1}{12}m\left(3(a^2+b^2)+h^2\right) \end{bmatrix}$$

Parallel axis theorem (Steiner theorem)

## Theorem

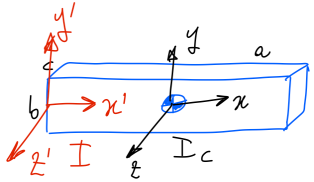
*Let $I_C$ be the inertia matrix with respect to a reference frame $\Sigma_C$ with origin on the center of mass. The inertia $I$ with respect to another reference frame $\Sigma$ obtained translating $\Sigma_C$ by the vector $r \in \mathbb{R}^3$ is given by*

$$I = I_C + mS^T(r)S(r) = I_C + m\left(r^T r\, I_{3\times 3} - rr^T\right)$$

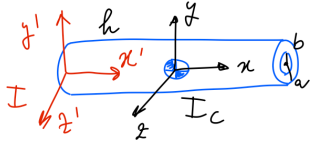
**Homework.** Prove

$$I_C + mS^T(r)S(r) = I_C + m\left(r^T r\, I_{3\times 3} - rr^T\right)$$

**Homework.**




$$I = I_C + m\left(\begin{bmatrix}-\frac{a}{2} & 0 & 0\end{bmatrix}\begin{bmatrix}-\frac{a}{2}\\0\\0\end{bmatrix} I_{3\times 3} - \begin{bmatrix}-\frac{a}{2}\\0\\0\end{bmatrix}\begin{bmatrix}-\frac{a}{2}\\0\\0\end{bmatrix}^T\right) = \cdots$$




$$I = I_C + m\left(\begin{bmatrix}-\frac{h}{2} & 0 & 0\end{bmatrix}\begin{bmatrix}-\frac{h}{2}\\0\\0\end{bmatrix} I_{3\times 3} - \begin{bmatrix}-\frac{h}{2}\\0\\0\end{bmatrix}\begin{bmatrix}-\frac{h}{2}\\0\\0\end{bmatrix}^T\right) = \cdots$$

# Kinetic Energy

Given $\mathcal{T}_{\ell_i} = \frac{1}{2} m_{\ell_i} \dot{p}_{\ell_i}^T \dot{p}_{\ell_i} + \frac{1}{2} \omega_i^T R_i \, I_{\ell_i}^i \, R_i^T \omega_i$, how can we compute

$$\dot{p}_{\ell_i}(q) = ? \qquad\qquad \omega_i(q) = ?$$

We know that the Cartesian velocity of the EE is related to the joint velocity via the Jacobian; however, this relationship holds also for intermediate links $i = 1, \dots, n$ (*Partial Jacobians*)

$$\dot{p}_{\ell_i} = J_P^{\ell_i}(q)\dot{q}, \qquad\qquad \omega_i = J_O^{\ell_i}(q)\dot{q}$$

$$\dot{p}_{\ell_i} = \begin{bmatrix} j_{P1}^{\ell_i} & j_{P2}^{\ell_i} & \cdots & j_{Pi}^{\ell_i} & 0 & \cdots & 0 \end{bmatrix} \begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \vdots \\ \dot{q}_i \\ \dot{q}_{i+1} \\ \vdots \\ \dot{q}_n \end{bmatrix}, \qquad \omega_i = \begin{bmatrix} j_{O1}^{\ell_i} & j_{O2}^{\ell_i} & \cdots & j_{Oi}^{\ell_i} & 0 & \cdots & 0 \end{bmatrix} \begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \vdots \\ \dot{q}_i \\ \dot{q}_{i+1} \\ \vdots \\ \dot{q}_n \end{bmatrix}$$

$$\dot{p}_{\ell_i} = \begin{bmatrix} j_{P1}^{\ell_i} & j_{P2}^{\ell_i} & \cdots & j_{Pi}^{\ell_i} & 0 & \cdots & 0 \end{bmatrix} \begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \vdots \\ \dot{q}_i \\ \dot{q}_{i+1} \\ \vdots \\ \dot{q}_n \end{bmatrix}, \qquad \omega_i = \begin{bmatrix} j_{O1}^{\ell_i} & j_{O2}^{\ell_i} & \cdots & j_{Oi}^{\ell_i} & 0 & \cdots & 0 \end{bmatrix} \begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \\ \vdots \\ \dot{q}_i \\ \dot{q}_{i+1} \\ \vdots \\ \dot{q}_n \end{bmatrix}$$

The columns of the Jacobians are

$$j_{Pj}^{\ell_i} = \begin{cases} z_{j-1}, & \text{prismatic joint} \\ z_{j-1} \times (p_{\ell_i} - p_{j-1}), & \text{revolute joint} \end{cases} \qquad j_{Oj}^{\ell_i} = \begin{cases} 0, & \text{prismatic joint} \\ z_{j-1}, & \text{revolute joint} \end{cases}$$

where

- $p_{j-1}$ is the position vector of the origin of Frame $\Sigma_{j-1}$ w.r.t. $\Sigma_0$
- $z_{j-1}$ is the unit vector of axis $z$ of Frame $\Sigma_{j-1}$ w.r.t. $\Sigma_0$

Finally

$$\mathcal{T}_{\ell_i} = \frac{1}{2} m_{\ell_i} \dot{q}^T (J_P^{\ell_i})^T J_P^{\ell_i} \dot{q} + \frac{1}{2} \dot{q}^T (J_O^{\ell_i})^T R_i \, I_{\ell_i}^i \, R_i^T J_O^{\ell_i} \dot{q}$$

where only the blue terms depend on $q$

$$\mathcal{T}_{\ell_i} = \frac{1}{2} m_{\ell_i} \dot{q}^T (J_P^{\ell_i})^T J_P^{\ell_i} \dot{q} + \frac{1}{2} \dot{q}^T (J_O^{\ell_i})^T R_i \, I_{\ell_i}^i \, R_i{}^T J_O^{\ell_i} \dot{q}$$

The **total Kinetic Energy** is a configuration-dependent quadratic function in $\dot{q}$

$$\begin{aligned} \mathcal{T}(q, \dot{q}) = \sum_{i=1}^{n} \mathcal{T}_{\ell_i} &= \frac{1}{2} \sum_{i=1}^{n} \left( m_{\ell_i} \dot{q}^T (J_P^{\ell_i})^T J_P^{\ell_i} \dot{q} + \dot{q}^T (J_O^{\ell_i})^T R_i \, I_{\ell_i}^i \, R_i^T J_O^{\ell_i} \dot{q} \right) \\ &= \frac{1}{2} \dot{q}^T \left[ \sum_{i=1}^{n} \left( m_{\ell_i} (J_P^{\ell_i})^T J_P^{\ell_i} + (J_O^{\ell_i})^T R_i \, I_{\ell_i}^i \, R_i^T J_O^{\ell_i} \right) \right] \dot{q} = \frac{1}{2} \dot{q}^T B(q) \dot{q} \end{aligned}$$

$$\mathcal{T}(q, \dot{q}) = \frac{1}{2} \dot{q}^T B(q) \dot{q}$$

where $B(q) \in \mathbb{R}^{n \times n}$ is the *inertia matrix* which is

- symmetric $B(q) = B(q)^T, \ \forall q \in \mathbb{R}^n$
- positive definite $B(q) \succ 0, \ \forall q \in \mathbb{R}^n \qquad (\Rightarrow$ nonsingular matrix $\forall q \in \mathbb{R}^n)$
- configuration-dependent

**Remarks**

1. $\mathcal{T}(q, \dot{q}) \geq 0$
2. $\mathcal{T}(q, \dot{q}) = 0$ if and only if $\dot{q} = 0$

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**Remarks**

1. $\mathcal{T}(q, \dot{q}) \geq 0$
2. $\mathcal{T}(q, \dot{q}) = 0$ if and only if $\dot{q} = 0$




These properties are the same for selecting a candidate Lyapunov function... it is not a coincidence and it will be exploited later!

**Assumption 1**: rigid links
**Assumption 2**: rigid transmission

The total Potential energy is given by the sum of the contributions relative to each link ($\mathcal{U}_{\ell_i}$) and the contributions relative to each joint motor actuator ($\mathcal{U}_{m_i}$)

$$\mathcal{U} = \sum_{i=1}^{n} (\mathcal{U}_{\ell_i} + \mathcal{U}_{m_i})$$

*From now on, we will consider only $\mathcal{U}_{\ell_i}$. The equations for $\mathcal{U}_{m_i}$ can be found in the textbook.*
Without elastic components, the potential energy is only due to the gravitational forces

$$\mathcal{U}_{\ell_i} = -\int_{V_{\ell_i}} g_0^T p_i^* \rho dV = -m_{\ell_i} g_0^T p_{\ell_i}, \qquad \mathcal{U} = -\sum_{i=1}^{n} m_{\ell_i} g_0^T p_{\ell_i}$$

where $g_0$ is the gravity acceleration vector in the base frame $\Sigma_0$ ($\mathbf{g}_0 = \begin{bmatrix} 0 \\ 0 \\ -g \end{bmatrix}$)

$$\mathcal{U} = \sum_{i=1}^{n} \mathcal{U}_i = -\sum_{i=1}^{n} m_{\ell_i} g_0^T p_{\ell_i}$$

The position of the center of mass of the link $i$ w.r.t. the base reference frame $\Sigma_0$, $p_{\ell_i}$, can be expressed w.r.t. the reference frame $\Sigma_i$ attached to the link, $p^i_{\ell_i}$, by

$$\begin{pmatrix} p_{\ell_i} \\ 1 \end{pmatrix} = T_1^0(q_1) T_2^1(q_2) \cdots T_i^{i-1}(q_i) \begin{pmatrix} p^i_{\ell_i} \\ 1 \end{pmatrix}$$

where $T_j^{j-1}(q_j)$ are the homogeneous transformation matrices.

- *The coordinate of the CoM with respect to $\Sigma_i$, $p^i_{\ell_i}$, is constant*
- *$\mathcal{U}_i = \mathcal{U}_i(q_1, q_2, \ldots, q_i)$ for open kinematic chain manipulators*

  $p_{\ell_1}$ is a function of $q_1$,

  $p_{\ell_2}$ is a function of $q_1, q_2$, ...

"link" causality


Project

# PROJECT – Assignment # 2

To do

- Compute the kinetic energy
- Compute the potential energy

# Equations of Motion

# Equations of Motion

The Lagrangian is given by

$$\mathcal{L}(q,\dot{q}) = \mathcal{T}(q,\dot{q}) - \mathcal{U}(q) = \frac{1}{2}\dot{q}^T B(q)\dot{q} + \sum_{i=1}^{n} m_{\ell_i} g_0^T p_{\ell_i}$$

We have to solve

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}}\right)^T - \left(\frac{\partial \mathcal{L}}{\partial q}\right)^T = \tau$$

where $\tau_i$ is the generalized force performing work on the $q_i$ generalized coordinate. $\tau_i$ is non-conservative.

Let's compute all the derivatives one by one

$$\left(\frac{\partial \mathcal{L}}{\partial \dot{q}}\right)^T = B(q)\dot{q}, \qquad \frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{q}}\right)^T = B(q)\ddot{q} + \dot{B}(q)\dot{q}$$

$$\left(\frac{\partial \mathcal{L}}{\partial q}\right)^T = \frac{1}{2}\left(\frac{\partial}{\partial q}\dot{q}^T B(q)\dot{q}\right)^T - \left(\frac{\partial \mathcal{U}}{\partial q}\right)^T$$

$$B(q)\ddot{q} + \dot{B}(q)\dot{q} - \frac{1}{2}\left(\frac{\partial}{\partial q}\dot{q}^T B(q)\dot{q}\right)^T + \left(\frac{\partial \mathcal{U}}{\partial q}\right)^T = \tau$$

For the $i-th$ DOF, we have

$$\sum_{j=1}^{n} b_{ij}(q)\ddot{q}_j + \sum_{j=1}^{n} \frac{db_{ij}(q)}{dt}\dot{q}_j - \frac{1}{2}\sum_{j=1}^{n}\sum_{k=1}^{n} \frac{\partial b_{jk}(q)}{\partial q_i}\dot{q}_k\dot{q}_j - \sum_{j=1}^{n} m_{\ell_j} g_0^T \frac{\partial p_{\ell_j}}{\partial q_i} = \tau_i$$

and finally

$$\sum_{j=1}^{n} b_{ij}(q)\ddot{q}_j + \underbrace{\sum_{j=1}^{n}\sum_{k=1}^{n} \frac{\partial b_{ij}(q)}{\partial q_k}\dot{q}_k\dot{q}_j - \frac{1}{2}\sum_{j=1}^{n}\sum_{k=1}^{n} \frac{\partial b_{jk}(q)}{\partial q_i}\dot{q}_k\dot{q}_j}_{\triangleq \sum_{j=1}^{n}\sum_{k=1}^{n} h_{ijk}(q)\dot{q}_k\dot{q}_j} - \underbrace{\sum_{j=1}^{n} m_{\ell_j} g_0^T j_{P_i}^{\ell_j}(q)}_{\triangleq g_i(q)} = \tau_i$$

$$\sum_{j=1}^{n} b_{ij}(q)\ddot{q}_j + \sum_{j=1}^{n}\sum_{k=1}^{n} h_{ijk}(q)\dot{q}_k\dot{q}_j + g_i(q) = \tau_i, \qquad i = 1, \ldots, n$$

- $b_{ii}(q)$ is the moment of inertia at Joint $i$ axis when the other joints are blocked ($q_j = const, \forall j \neq i$)
- $b_{ii}(q) = b_{ii} > 0$
- $b_{ij}$ effects of acceleration of Joint $j$ on Joint $i$

$$\sum_{j=1}^{n} b_{ij}(q)\ddot{q}_j + \sum_{j=1}^{n}\sum_{k=1}^{n} h_{ijk}(q)\dot{q}_k\dot{q}_j + g_i(q) = \tau_i, \qquad i = 1, \ldots, n$$

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- $b_{ii}(q) = b_{ii} > 0$
- $b_{ij}$ effects of acceleration of Joint $j$ on Joint $i$
- $h_{iij}\dot{q}_j^2$ is the centrifugal effect induced on Joint $i$ by velocity of Joint $j$, ($h_{iii} \equiv 0, \forall i$)
- $h_{ijk}\dot{q}_j\dot{q}_k$ is the Coriolis effect induced on Joint $i$ by velocities of Joints $j$ and $k$

# Equations of Motion

$$\sum_{j=1}^{n} b_{ij}(q)\ddot{q}_j + \sum_{j=1}^{n}\sum_{k=1}^{n} h_{ijk}(q)\dot{q}_k\dot{q}_j + g_i(q) = \tau_i, \qquad i = 1, \dots, n$$

- $b_{ii}(q)$ is the moment of inertia at Joint $i$ axis when the other joints are blocked ($q_j = const, \forall j \neq i$)
- $b_{ii}(q) = b_{ii} > 0$
- $b_{ij}$ effects of acceleration of Joint $j$ on Joint $i$
- $h_{iji}\dot{q}_j^2$ is the centrifugal effect induced on Joint $i$ by velocity of Joint $j$, ($h_{iii} \equiv 0, \forall i$)
- $h_{ijk}\dot{q}_j\dot{q}_k$ is the Coriolis effect induced on Joint $i$ by velocities of Joints $j$ and $k$
- $g_i$ is the moment generated at Joint $i$ axis of the manipulator by gravity.

# Equations of Motion

$$\sum_{j=1}^{n} b_{ij}(q)\ddot{q}_j + \sum_{j=1}^{n}\sum_{k=1}^{n} h_{ijk}(q)\dot{q}_k\dot{q}_j + g_i(q) = \tau_i, \qquad i = 1, \ldots, n$$

- linear terms in acceleration $\ddot{q}$
- quadratic terms in velocity $\dot{q}$
- nonlinear terms in position $q$

# Equations of Motion

The equations of motion

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can be rewritten as

$$\sum_{j=1}^{n} b_{ij}(q)\ddot{q}_j + \sum_{j=1}^{n} c_{ij}(q, \dot{q})\dot{q}_j + g_i(q) = \tau_i, \qquad i = 1, \ldots, n$$

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The choice of $\{c_{ij}\}$ is not unique!

# Equations of Motion

The equations of motion

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The choice of $\{c_{ij}\}$ is not unique!

However, there is a clever choice: *Christoffel symbols of the first type*

# Christoffel symbols of the first type

$$
\begin{aligned}
\sum_{j=1}^{n} c_{ij}(q)\dot{q}_j &= \sum_{j=1}^{n}\sum_{k=1}^{n} h_{ijk}\dot{q}_k\dot{q}_j = \sum_{j=1}^{n}\sum_{k=1}^{n}\left(\frac{\partial b_{ij}}{\partial q_k} - \frac{1}{2}\frac{\partial b_{ij}}{\partial q_i}\right)\dot{q}_k\dot{q}_j \\
&= \frac{1}{2}\sum_{j=1}^{n}\sum_{k=1}^{n}\frac{\partial b_{ij}}{\partial q_k}\dot{q}_k\dot{q}_j + \frac{1}{2}\sum_{j=1}^{n}\sum_{k=1}^{n}\left(\frac{\partial b_{ik}}{\partial q_j} - \frac{\partial b_{jk}}{\partial q_i}\right)\dot{q}_k\dot{q}_j \\
&= \sum_{j=1}^{n}\sum_{k=1}^{n}\underbrace{\frac{1}{2}\left(\frac{\partial b_{ij}}{\partial q_k} + \frac{\partial b_{ik}}{\partial q_j} - \frac{\partial b_{jk}}{\partial q_i}\right)}_{\triangleq c_{ijk}}\dot{q}_k\dot{q}_j \\
&= \sum_{j=1}^{n}\underbrace{\sum_{k=1}^{n} c_{ijk}\dot{q}_k}_{\triangleq c_{ij}}\dot{q}_j
\end{aligned}
$$

Property: $c_{ijk} = c_{ikj}$

The $n$ equations of motion

$$\sum_{j=1}^{n} b_{ij}(q)\ddot{q}_j + \sum_{j=1}^{n} c_{ij}(q,\dot{q})\dot{q}_j + g_i(q) = \tau_i, \qquad i = 1, \dots, n$$

can be written more compactly as

$$B(q)\ddot{q} + C(q,\dot{q})\dot{q} + g(q) = \tau$$

If we have to take into account friction (*viscous friction* $F_v\dot{q}$, *Coulomb friction* $F_s\text{sign}(\dot{q})$)

$$B(q)\ddot{q} + C(q,\dot{q})\dot{q} + F_v\dot{q} + F_s\text{sign}(\dot{q}) + g(q) = \tau$$

If the end-effector interacts with the environment via the *external wrench* $h_e$ we end up with

$$B(q)\ddot{q} + C(q,\dot{q})\dot{q} + F_v\dot{q} + F_s\text{sign}(\dot{q}) + g(q) = \tau - J^T(q)h_e$$

Set of $n$ nonlinear second-order differential equations.


Project

# PROJECT – Assignment # 3

To do

- equations of motion (dynamic model)